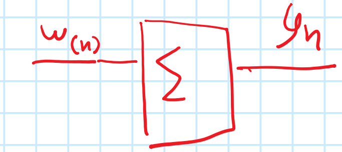
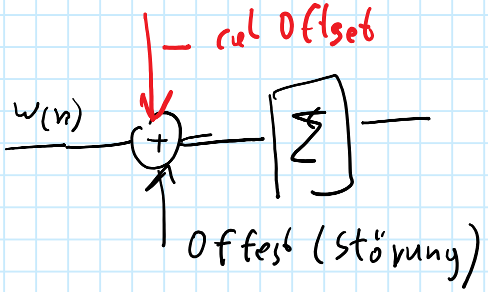
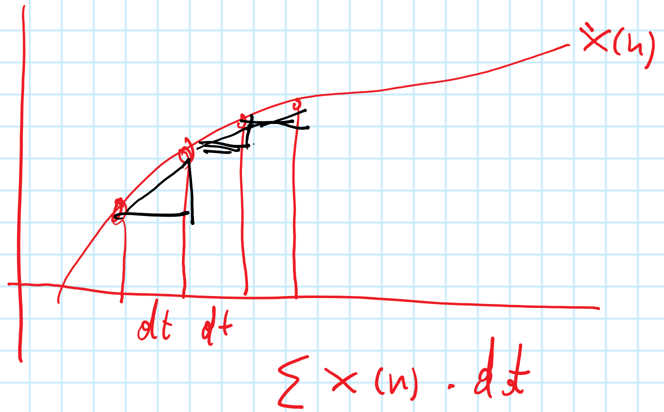
1 Gyrosensor als Lagesensor

Gyroskop

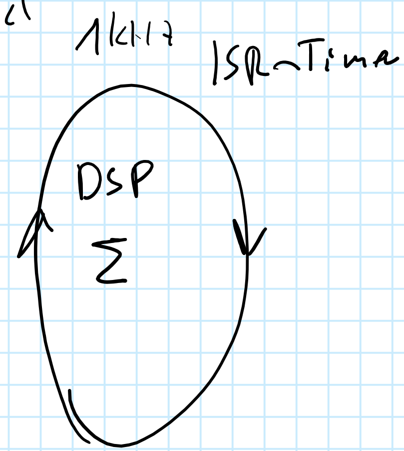


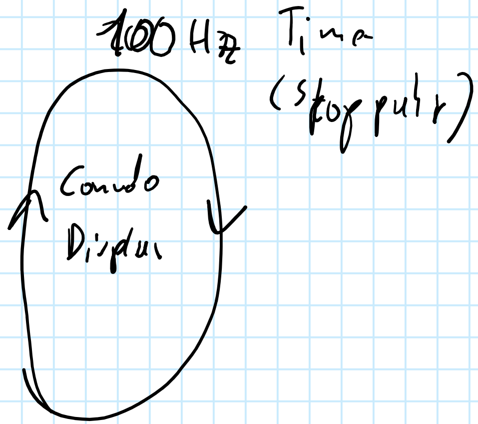


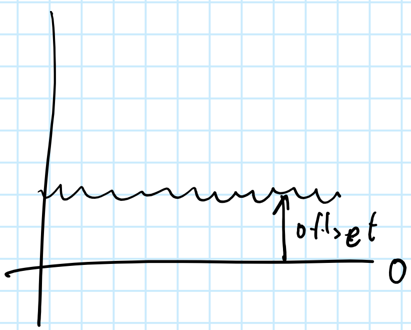


– Offset abgleich

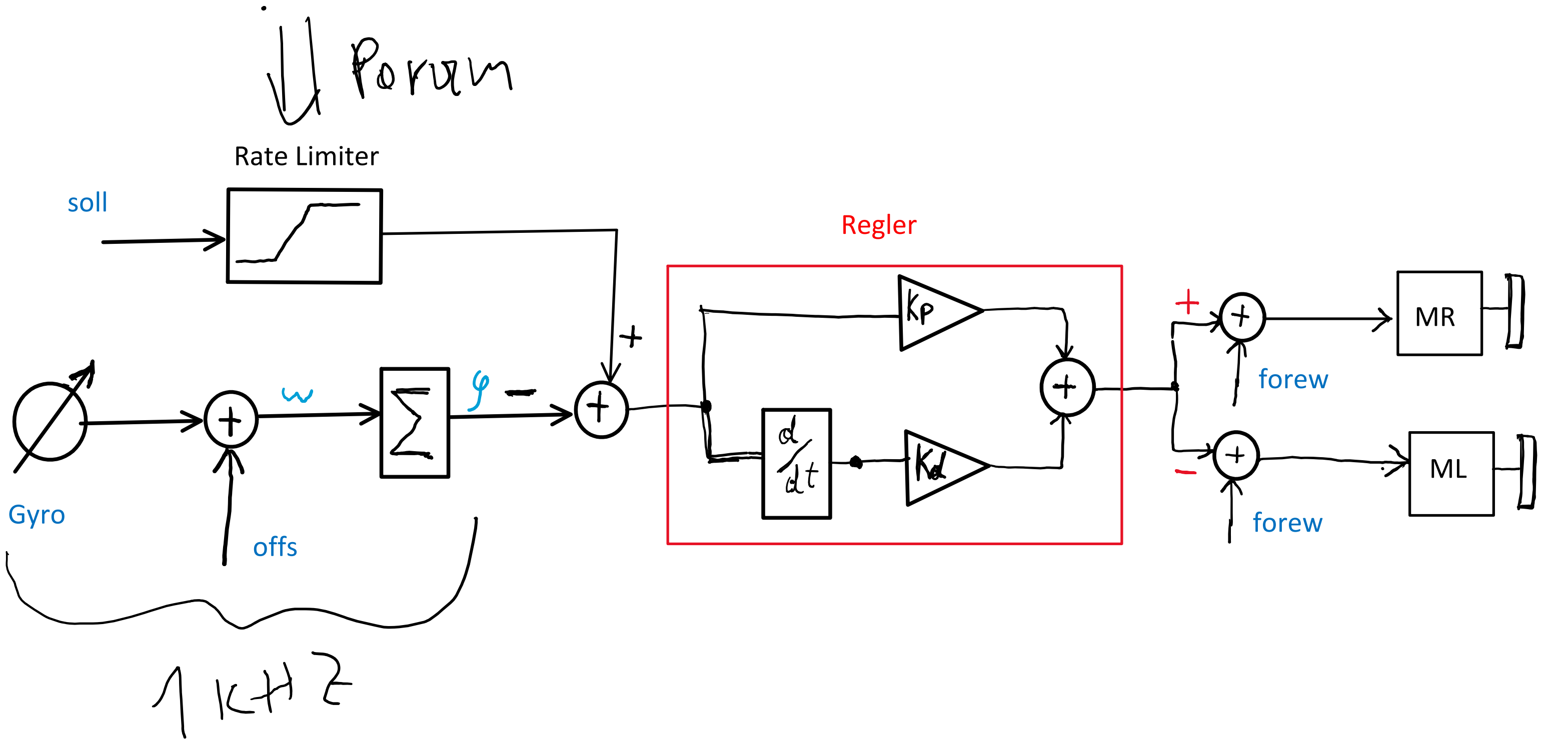
– Reset







2 GyroControl



3 Code Snippeds

```
#define CHECK_CANCEL() if( drCmd.doCancel() ) goto StopAll

TTMailBox drCmd;
bool calFlag = false;
```

```
else if (cmd == 10) { // IPC msg to DriveTask
  drCmd.send(ua0.ReadI16());
  // DriveTask takes Parameters from SvMessage
  vTaskDelay(5);
}
```

```
void DriveTask(void* arg)
{
  int cmd;
  while (1) {
    cmd = drCmd.recv();
    if (cmd == 1) {
      StraightAndCurve(ua0.ReadF(), ua0.ReadI16());
      ua0.SvMessage("Curve finished!!");
    }
  }
}
```

```
void StraightAndCurve(float forew, int dist)
{
  ua0.SvPrintf("drive %1.2f %d", forew, dist);
  MyDelay(500);
  while(1) {
    ua0.SvMessage("Straight");
    integ.Reset(); rgl.setDemand2(0); // no ramp
    rgl.on = 1; encL.cnt = 0;
    rgl.forew = forew;
    while (encL.absCnt() < dist) {
      CHECK_CANCEL();
      vTaskDelay(1);
    }
    ua0.SvMessage("Curve");
    integ.Reset(); rgl.setDemand(1800); // with ramp
    while (rgl.getActDemand() < 1800.0) {
      CHECK_CANCEL();
      vTaskDelay(1);
    }
  }
StopAll:
  drCmd.set(0); // encL.cnt = 0;
  rgl.on=0; vTaskDelay(2);
  motL.setPow2(0); motR.setPow2(0);
  integ.Reset(); rgl.setDemand2(0);
}
```

4 xxxx

-

5 xxxx

-